Nonlinear Primal-Dual

Anonymous Author(s)
Affiliation
Address
email

Abstract

In this paper, we

1 Introduction

In this paper, we consider a structured nonlinear composite optimization problem

$$\min_{\boldsymbol{x}\in\mathcal{X}} F(\boldsymbol{x}) := f(\boldsymbol{x}) + g(\boldsymbol{x}) + h\big(\boldsymbol{c}(\boldsymbol{x})\big), \tag{1.1}$$

where $\mathcal{X} \subseteq \mathbb{R}^n$ is a nonempty closed subset of $\mathbb{R}^n$ which admits an easy projection, $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable (possibly nonconvex), $g : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a proper lower semicontinuous and prox-bounded function relative to its domain $\operatorname{dom} g$ (possibly nonconvex), $h : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$ is proper lower semicontinuous (possibly nonsmooth and nonconvex), and $\boldsymbol{c} : \mathbb{R}^n \to \mathbb{R}^m$ is a C^1 mapping. Moreover, we assume that the problem is well-posed, i.e., the feasible set is nonempty and the optimal value is finite.

Due to the separable structure of the objective function, operator-splitting approaches have emerged as a standard and effective technique for addressing problem (1.1). For the general composite setting, a typical reformulation introduces an auxiliary variable $y = c(\boldsymbol{x})$, leading to a block-separable formulation:

$$\min_{\boldsymbol{x}\in\mathcal{X},\, \boldsymbol{y}\in\mathbb{R}^m} f(\boldsymbol{x}) + g(\boldsymbol{x}) + h(\boldsymbol{y}), \quad \text{s.t.} \quad \boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{0}.$$

Motivating Examples

A special case of Problem (1.1) is obtained by taking $h = \delta_{\{\boldsymbol{0}\}}$, which yields the nonlinear equality-constrained problem

$$\min_{\boldsymbol{x}\in\mathcal{X}} f(\boldsymbol{x}) + g(\boldsymbol{x}), \quad \text{s.t.} \quad \boldsymbol{c}(\boldsymbol{x}) = \boldsymbol{0}.$$

1.1 Related work

Nonlinear Constrained Optimization.

Noconvex Primal-Dual Method. Not limited in the ADMM-related work

Nonsmooth Compositional Optimization. In nonsmooth compositional optimization, such as Problem (1.1), significant work has been done in the case where $h \circ \boldsymbol{c}$ is either convex [14, 10] or weakly convex [7, 5, 6, 11, 9]. Specifically, prior works [7, 5, 6, 11] typically ensure weak convexity of $h \circ \boldsymbol{c}$ by imposing convexity on h and smoothness on $\boldsymbol{c}$. Recent work [9] assumes both h and $\boldsymbol{c}$ are weakly convex with h being component-wise non-decreasing, thereby ensuring the weak convexity of $h \circ \boldsymbol{c}$. However, when h itself is nonconvex, as in our proposal, the composite function $h \circ \boldsymbol{c}$ is not necessarily weakly convex, rendering these methods inapplicable. For the fully nonconvex case, [1] reformulates the problem as constrained optimization and establishes asymptotic convergence

Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

via a Lagrangian-based method. Under the assumption that $\boldsymbol{c}$ is linear, some studies [2, 3, 4, 8] have proposed full splitting algorithms and established global convergence guarantees in a single-loop framework. Recently, [12] considers a similar setting to Problem (1.1) with nonlinear $\boldsymbol{c}$. Using a smoothing technique, they provide non-asymptotic convergence analysis, while [13] systematically analyzed scenarios where h is Lipschitz continuous or non-Lipschitz continuous, establishing non-asymptotic and subsequential convergence results.

Contribution

1. the constant penalty parameter for nonconvex ADMM;

2. the $\mathcal{O}(\varepsilon^{-?})$ between the $\mathcal{O}(\varepsilon^{-2})$ bound for smooth bound and the $\mathcal{O}(\varepsilon^{-4})$ bound for the subgradient method.

Notation

$\mathbb{R}^n$ denotes the n-dimensional Euclidean space, and $\mathbb{R}^n_+$ represents the non-negative orthant. We use $[m]$ to denote the set $\{1, \ldots, m\}$. The inner product between two vectors is denoted by $\langle \cdot, \cdot \rangle$, and the Euclidean norm of matrices and vectors is denoted by $\|\cdot\|$. The distance function between a vector $\mathbf{x}$ and a set $\mathcal{X} \subseteq \mathbb{R}^n$ is defined as $\operatorname{dist}(\mathbf{x}, \mathcal{X}) := \inf_{\mathbf{y} \in \mathcal{X}} \|\mathbf{y} - \mathbf{x}\|$. For a proper extended real-valued function r, its *domain* is defined as $\operatorname{dom}(r) := \{\mathbf{x} \in \mathbb{R}^n : r(\mathbf{x}) < +\infty\}$. A function r is considered *proper* if $\operatorname{dom}(r) \neq \emptyset$ and it does not take the value $-\infty$. It is *closed* if it is lower semicontinuous, meaning $\liminf_{\mathbf{x} \to \mathbf{x}^0} r(\mathbf{x}) \geq r(\mathbf{x}^0)$ for any $\mathbf{x}^0 \in \mathbb{R}^n$. For a convex function r at $\mathbf{x}$, its *subgradient* is denoted by $\partial r(\mathbf{x}) := \{\mathbf{d} \in \mathbb{R}^n : r(\mathbf{y}) \geq r(\mathbf{x}) + \langle \mathbf{d}, \mathbf{y} - \mathbf{x} \rangle, \forall \mathbf{y} \in \mathbb{R}^n,\ \mathbf{x} \in \operatorname{dom}(r)\}$. The *proximal map* associated with a proper, closed, and convex function $r : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ at $\mathbf{x} \in \mathbb{R}^n$ with $\eta > 0$ is uniquely defined by $\operatorname{prox}_{\eta r}(\mathbf{x}) = \operatorname{argmin}_{\mathbf{y} \in \mathbb{R}^n} \left\{ r(\mathbf{y}) + \frac{1}{2\eta} \|\mathbf{x} - \mathbf{y}\|^2 \right\}$.

2 Preliminaries

Definition 1 (KKT Point and $(\varepsilon_1, \varepsilon_2)$-KKT Point) *A point $\boldsymbol{x}^* \in \mathcal{X}$ is called a* KKT point *of problem (1.1) if*

$$0 \in \nabla f(\boldsymbol{x}^*) + \partial g(\boldsymbol{x}^*) + J_{\boldsymbol{c}}(\boldsymbol{x}^*)^\top \partial h\big(\boldsymbol{c}(\boldsymbol{x}^*)\big) + \mathcal{N}_{\mathcal{X}}(\boldsymbol{x}^*). \tag{2.1}$$

Moreover, for $\varepsilon_1, \varepsilon_2 > 0$, a point $\boldsymbol{x}^* \in \mathcal{X}$ is called an $(\varepsilon_1, \varepsilon_2)$-KKT point *if there exists $\boldsymbol{y}^* \in \mathbb{R}^m$ such that:*

$$\operatorname{dist}\Big(-\mathcal{N}_{\mathcal{X}}(\boldsymbol{x}^*),\ \nabla f(\boldsymbol{x}^*) + \partial g(\boldsymbol{x}^*) + J_{\boldsymbol{c}}(\boldsymbol{x}^*)^\top \partial h\big(\boldsymbol{y}^*\big) \Big) \leq \varepsilon_1, \tag{2.2}$$

$$\|\boldsymbol{c}(\boldsymbol{x}^*) - \boldsymbol{y}^*\| \leq \varepsilon_2. \tag{2.3}$$

Remark 2.1 *Condition (2.1) is derived via the chain rule under Assumption 1. In the block-separable formulation (**??**), introducing a dual multiplier $\lambda^* \in \mathbb{R}^m$, the KKT condition (2.1) is equivalently expressed as:*

$$0 \in \nabla f(x^*) + \partial g(x^*) + J_{\boldsymbol{c}}(x^*)^T \lambda^*, \tag{2.4}$$

$$0 \in \partial h(y^*) - \lambda^*, \tag{2.5}$$

$$0 = \boldsymbol{c}(x^*) - y^*, \tag{2.6}$$

which reduces to (2.1) upon eliminating λ^*.

2.1 Main Assumptions

To establish the ensuing analysis, we introduce the following standard assumptions for problem

Assumption 1 (Constraint Qualification) *The following constraint qualification holds for problem (1.1):*

$$\textit{[CQ]} \qquad \nabla \boldsymbol{c}(\boldsymbol{x})^T y = 0, \quad y \in \partial^\infty h\big(\boldsymbol{c}(\boldsymbol{x})\big) \implies y = 0.$$

3 Methodology

To decouple the nonlinear term $h \circ \boldsymbol{c}$, we introduce an auxiliary variable $\boldsymbol{y}$ and a residual variable $\boldsymbol{z}$ and rewrite (1.1) as

$$\min_{\boldsymbol{x},\boldsymbol{y},\boldsymbol{z}} \; f(\boldsymbol{x}) + g(\boldsymbol{x}) + h(\boldsymbol{y}) \quad \text{s.t.} \quad \boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{y} = \boldsymbol{z}, \;\; \boldsymbol{z} = \boldsymbol{0}. \tag{3.1}$$

Following [], we define the perturbed proximal Lagrangian

$$\mathcal{L}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := f(\boldsymbol{x}) + g(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \, \boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{y} - \boldsymbol{z} \rangle + \langle \boldsymbol{\mu}, \, \boldsymbol{z} \rangle + \frac{\alpha}{2}\|\boldsymbol{z}\|^2 - \frac{\beta}{2}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2 + \frac{\rho}{2}\|\boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{y}\|^2$$

where $\boldsymbol{\lambda}, \boldsymbol{\mu} \in \mathbb{R}^m$ are dual variables associated with the constraints in (3.1). The parameter α plays the role of a primal penalty on $\boldsymbol{z}$, while β introduces a dual regularization term.

The proximal Lagrangian function, obtained by eliminating $\boldsymbol{z}$ via its stationary condition $\boldsymbol{z}^*(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \frac{\boldsymbol{\lambda} - \boldsymbol{\mu}}{\alpha}$, reduces to

$$\mathcal{L}_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}, \boldsymbol{\mu}) := l_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}) - \frac{1}{2\rho}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2, \tag{3.2}$$

where $l_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) + g(\boldsymbol{x}) + h(\boldsymbol{y}) + \langle \boldsymbol{\lambda}, \, \boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{y} \rangle + \frac{\rho}{2}\|\boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{y}\|^2$ and $\rho > 0$ is defined as

$$\rho := \frac{\alpha}{1 + \alpha\beta} > 0. \tag{3.3}$$

Remark 3.1 *Here,*

$$\phi_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\mu}) = \max_{\boldsymbol{\lambda}} \left\{ l_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\lambda}) - \frac{1}{2\rho}\|\boldsymbol{\lambda} - \boldsymbol{\mu}\|^2 \right\}$$

therefore

$$\max_{\boldsymbol{\mu}} \min_{\boldsymbol{x},\boldsymbol{y}} \phi_\rho(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\mu})$$

but the update of $\boldsymbol{\lambda}$ is not considered.

3.1 Algorithm

We now describe one iteration of the proposed method. Given $(\boldsymbol{x}^k, \boldsymbol{y}^k, \boldsymbol{z}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)$, the blocks are updated as follows.

Update of $\boldsymbol{x}$. We apply a prox-linear / proximal-gradient step to the x-block subproblem on l_ρ

$$\boldsymbol{x}^{k+1} \in \operatorname*{argmin}_{\boldsymbol{x} \in \mathcal{X}} \left\{ g(\boldsymbol{x}) + \left\langle \nabla f(\boldsymbol{x}^k) + J_{\boldsymbol{c}}(\boldsymbol{x}^k)^\top \left(\boldsymbol{\lambda}^k + \rho(\boldsymbol{c}(\boldsymbol{x}^k) - \boldsymbol{y}^k) \right), \; \boldsymbol{x} - \boldsymbol{x}^k \right\rangle + \frac{1}{2\eta}\|\boldsymbol{x} - \boldsymbol{x}^k\|^2 \right\}. \tag{3.4}$$

Update of $\boldsymbol{y}$. The y-block is updated by a proximal gradient step on h:

$$\boldsymbol{y}^{k+1} \in \operatorname*{argmin}_{\boldsymbol{y} \in \mathbb{R}^m} \left\{ h(\boldsymbol{y}) + \langle -\boldsymbol{\lambda}^k + \rho\,(\boldsymbol{y} - \boldsymbol{c}(x^{k+1}), \boldsymbol{y} - \boldsymbol{y}^k \rangle + \frac{1}{2\tau}\|\boldsymbol{y} - \boldsymbol{y}^k\|^2 \right\} \tag{3.5}$$

Update of $\boldsymbol{\mu}$. The auxiliary multiplier $\boldsymbol{\mu}$ is updated via a gradient acent step

$$\boldsymbol{\mu}^{k+1} = \boldsymbol{\mu}^k + \theta_k \left(\boldsymbol{\lambda}^k - \boldsymbol{\mu}^k \right), \qquad \theta_k \in (0, 1]. \tag{3.6}$$

Update of $\boldsymbol{\lambda}$. The multiplier $\boldsymbol{\lambda}$ is updated by the unique maximizer of $\mathcal{L}_\rho$ with respect to $\boldsymbol{\lambda}$:

$$\boldsymbol{\lambda}^{k+1} = \boldsymbol{\mu}^{k+1} + \rho\big(\boldsymbol{c}(\boldsymbol{x}^{k+1}) - \boldsymbol{y}^{k+1}\big). \tag{3.7}$$

Update of z. Finally, z admits a closed-form expression from the optimality condition of (3.2) with respect to z:

$$z^{k+1} = \frac{1}{\alpha}(\lambda^{k+1} - \mu^{k+1}). \tag{3.8}$$

Remark 3.2

Algorithm 1 Peturbed linearized ADMM

1: **Input:** initial point $(x^0, y^0, \lambda^0, \mu^0)$, parameters $\eta > 0$, $\tau > 0$, $\alpha > 0$, $\beta > 0$, damping sequence $\{\theta_k\}_{k\geq 0}$ and ρ defined in (3.3).
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Update x^{k+1} by (3.4).
4: Update y^{k+1} by (3.5).
5: Update μ^{k+1} by (3.6) with $\theta_k = \frac{\delta_k}{1+\|\lambda^k - \mu^k\|^2}$.
6: Update λ^{k+1} by (3.7).
7: Update z^{k+1} by (3.8).
8: **end for**
9: **Output:** an iterate x^R (e.g., the randomly selected one).

4 Convergence Analysis

Lemma 1 *Let $\{\mu_k\}$ and $\{\lambda_k\}$ be the sequences generated by Algorithm (1). Then, we have*

$$\|\mu_{k+1} - \mu_k\|^2 \leq \frac{\delta_k^2}{4}, \qquad \|\lambda_k - \mu_k\|^2 \leq \frac{\delta_k}{\tau_k}, \tag{4.1}$$

and

$$\|\lambda_{k+1} - \lambda_k\|^2 \leq 2\|\lambda_{k+1} - \mu_{k+1}\|^2 + 2\|\mu_{k+1} - \lambda_k\|^2. \tag{4.2}$$

Lemma 2 *Under Assumptions, let $\{(x^k, y^k, \lambda^k, \mu^k)\}$ be the sequence generated by Algorithm 1. Then for any $k \geq 1$,*

$$\|\mu^{k+1}\| \leq \|\mu^0\| + \frac{1}{2}\sum_{j=0}^{\infty}\theta_k, \tag{4.3}$$

and

5 Conclusion

6 Limitation

References

[1] Jérôme Bolte, Shoham Sabach, and Marc Teboulle. Nonconvex lagrangian-based optimization: Monitoring schemes and global convergence. *Mathematics of operations research*, 43(4):1210–1232, 2018.

[2] Radu Ioan Boţ, Ernö Robert Csetnek, and Szilárd Csaba László. An inertial forward–backward algorithm for the minimization of the sum of two nonconvex functions. *EURO Journal on Computational Optimization*, 4(1):3–25, 2016.

[3] Radu Ioan Bot, Erno Robert Csetnek, and Dang-Khoa Nguyen. A proximal minimization algorithm for structured nonconvex and nonsmooth problems. *SIAM Journal on Optimization*, 29(2):1300–1328, 2019.

[4] Radu Ioan Boţ and Dang-Khoa Nguyen. The proximal alternating direction method of multipliers in the nonconvex setting: convergence analysis and rates. *Mathematics of Operations Research*, 45(2):682–712, 2020.

[5] Damek Davis and Dmitriy Drusvyatskiy. Stochastic model-based minimization of weakly convex functions. *SIAM Journal on Optimization*, 29(1):207–239, 2019.

[6] D. Drusvyatskiy and C. Paquette. Efficiency of minimizing compositions of convex functions and smooth maps. *Mathematical Programming*, 178(1):503–558, Nov 2019.

[7] John C. Duchi and Feng Ruan. Stochastic methods for composite and weakly convex optimization problems. *SIAM Journal on Optimization*, 28(4):3229–3259, 2018.

[8] Jiahong Guo, Xiao Wang, and Xiantao Xiao. Preconditioned primal-dual gradient methods for nonconvex composite and finite-sum optimization. *arXiv preprint arXiv:2309.13416*, 2023.

[9] Quanqi Hu, Dixian Zhu, and Tianbao Yang. Non-smooth weakly-convex finite-sum coupled compositional optimization. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 5348–5403. Curran Associates, Inc., 2023.

[10] Bokun Wang and Tianbao Yang. Alexr: An optimal single-loop algorithm for convex finite-sum coupled compositional stochastic optimization, 2024.

[11] Xiao Wang and Xiaojun Chen. Complexity of finite-sum optimization with nonsmooth composite functions and non-lipschitz regularization. *SIAM Journal on Optimization*, 34(3):2472–2502, 2024.

[12] Y. Yao, Q. Lin, and T. Yang. A note on complexity for two classes of structured non-smooth non-convex compositional optimization. *arXiv preprint arXiv:2411.14342*, 2024.

[13] H. Zhang, N. Marumo, T. K. Pong, and A. Takeda. Complexity and convergence analysis of a single-loop sdcam for lipschitz composite optimization and beyond. *Manuscript*, 2025. Preprint.

[14] Zhe Zhang and Guanghui Lan. Optimal algorithms for convex nested stochastic composite optimization. *arXiv preprint arXiv:2011.10076*, 2020.